CHINOOK MUSIC STORE DATA ANALYSIS

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1. INTRODUCTION

The purpose of this project is to analyze sales data from Chinook music store using data analysis and business intelligence techniques. The project focuses on transforming raw transactional data into meaningful insights that help us understand sales performance, customer behavior, and music catalog characteristics.

To achieve this objective, the data was processed and analyzed using Power BI, supported by SQL, Power Query, and DAX measures. A multi-page interactive dashboard was developed to provide different analytical perspectives, including an executive overview, artist performance analysis, catalog exploration, customer insights, and time-based revenue trends.

The dashboard enables users to explore key business metrics such as total revenue, customer activity, top performing artists, and revenue growth over time. Interactive slicers and dynamic visualizations allow users to filter and analyze the data from multiple dimensions such as genre, artist, country, and year.

The final solution provides a structured analytical tool that helps transform raw data into actionable insights through interactive visual analytics.

Tools Used:

- **SQL**: Used to create and manage the dataset.
- **Power BI**: Used for data modeling, visualization, and dashboard creation.
- **Power Query**: Used for data cleaning and transformation.
- **DAX**: Used to create calculated measures.
- **Figma**: Used to design the dashboard layout and visual style.

2. DATASET DESCRIPTION

The dataset used in this project represents the sales operations of Chinook music store. It contains information about artists, albums, music tracks, customers, employees, and sales transactions.

The data allows analysis of music sales performance, customer purchasing behavior, and the structure of the music catalog.

The dataset consists of several related tables, each representing a different aspect of the business.

Main Tables

Table	Description
Artist	Contains information about music artists.
Album	Contains albums created by artists. Each album belongs to one artist.
Track	Contains detailed information about individual music tracks including genre, media type, and duration.
Customer	Contains customer information such as name, country, and contact details.
Employee	Contains information about employees who act as customer support representatives.
Invoice	Represents customer purchases and contains order-level information such as invoice date and customer ID.
InvoiceLine	Contains individual items purchased within each invoice, including track ID, quantity, and unit price.
Genre	Contains music genre categories used to classify tracks.
MediaType	Describes the format of the track such as MPEG audio file.

The dataset contains transactional sales data where each invoice represents a purchase made by a customer, and each invoice line represents a specific track purchased in that transaction.

This structure allows the analysis of revenue, track popularity, artist performance, and customer purchasing patterns.

3. DATA MODELING

Data Model Design

To enable efficient analysis and reporting, the dataset was reorganized into a model centered around a **fact table and several supporting dimension tables**.

The **InvoiceLine table** was selected as the central **fact table**, as it represents the lowest level of sales transactions. Each record in this table corresponds to a specific track purchased within an invoice.

Several descriptive tables were transformed into **dimension tables**, providing attributes that allow the data to be analyzed from multiple perspectives such as artist, genre, customer, and time.

Fact Table

Table	Description
InvoiceLine	The central fact table containing transactional sales records. Each row represents a purchased track within a specific invoice and includes quantity and unit price information.

The fact table was used to calculate key business metrics such as:

- Total Revenue
- Tracks Sold
- Average Order Value
- Revenue by Artist
- Revenue by Genre

Dimension Tables

Dimension tables provide descriptive information used to categorize and filter the fact data.

Dimension Table	Description
Dim_Customer	Contains customer information such as name, location, and contact details.
Dim_Track	Contains track-level information including track name, album title, artist name, genre, media type, and duration.
DimDate	A generated date table used for time-based analysis such as yearly or monthly trends.
Employee	Contains information about employees including customer support representatives.
Invoice	Contains order-level information including invoice date and customer ID.

To simplify analysis, several normalized tables from the original dataset (Artist, Album, Genre, and MediaType) were combined into the **Dim_Track dimension**. This denormalization step improves usability and reduces the number of relationships required in the analytical model.

Relationships

The data model is organized around the **InvoiceLine fact table**, which connects to other tables through key relationships. The relationships allow Power BI to aggregate and filter transactional data across multiple business dimensions.

For example:

- Tracks purchased in each transaction are linked through the **TrackID** field.
- Customer purchases are linked through the **Invoice and Customer tables**.
- Time-based analysis is enabled through the **DimDate table** connected to invoice dates.

This structure allows users to analyze sales performance across different perspectives including:

- Artist performance
- Genre popularity
- Customer purchasing behavior
- Geographic distribution of customers
- Revenue trends over time

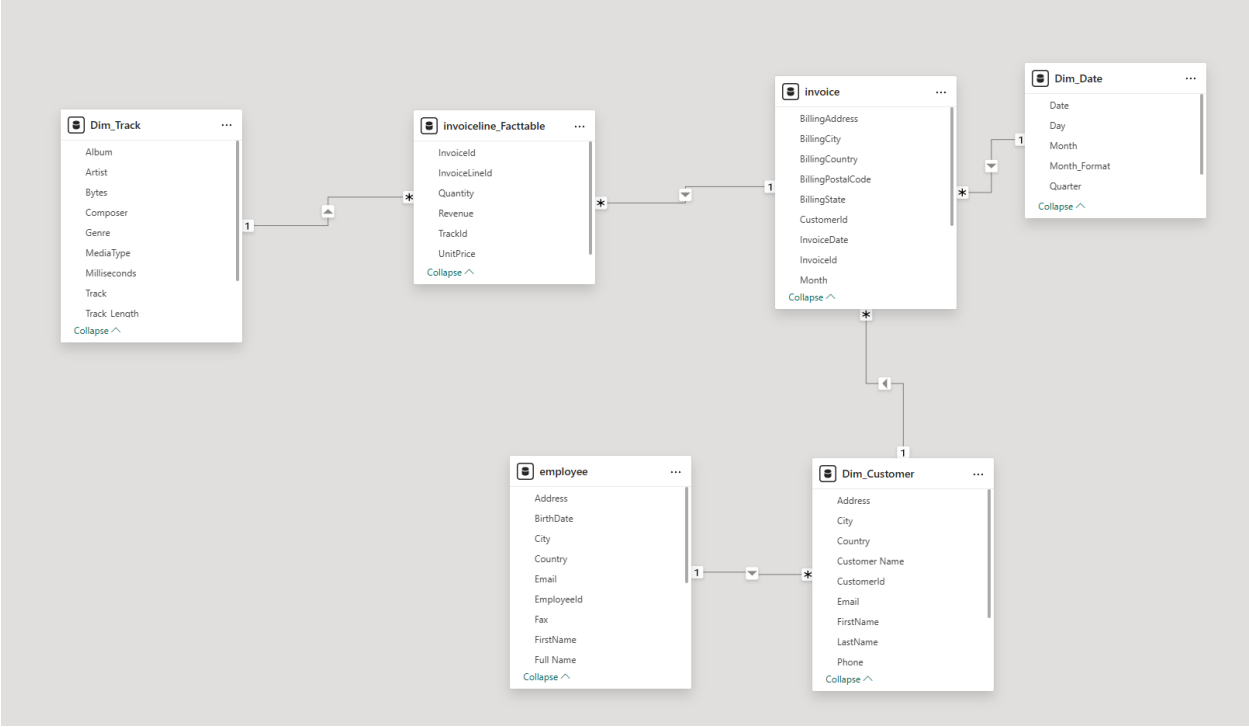


Figure: Power BI data model showing relationships between fact and dimension tables.

DEMO. DATA PREPARATION

Data Preparation Process

Before building the analytical model and dashboard, several data preparation steps were performed using **Power Query in Power BI**. These transformations were necessary to ensure that the dataset was clean, properly structured, and optimized for analysis.

Power Query was used to load the tables, perform transformations, and create additional fields required for business analysis.

Data Cleaning and Transformation

Several transformations were applied to improve data usability and consistency.

Key steps included:

• Adjusting column data types to ensure correct calculations

• Renaming tables and columns for better readability

• Removing unnecessary tables and columns from the dataset

• Creating additional calculated columns for analysis

These transformations ensured that the dataset could be efficiently used for reporting and analysis within Power BI.

Creation of Revenue Column

A new column called **Revenue** was created in the *InvoiceLine* table using Power Query.

This column calculates the revenue generated from each purchased track.

Formula: Revenue = UnitPrice * Quantity

Creating this column simplifies revenue calculations and allows revenue to be easily aggregated across different dimensions such as artist, genre, customer, and country.

Creation of Dim_Track Table

To simplify analysis, several normalized tables from the original dataset were combined into a single **Dim_Track dimension table**.

The following tables were merged:

Track

Album

Artist

Genre

MediaType

This transformation allowed all track-related descriptive attributes to be accessed from one table, reducing complexity in the data model and making it easier to build visualizations.

The resulting **Dim_Track table** includes attributes such as:

- Track name
- Album title
- Artist name
- Genre
- Media type
- Track duration

Creation of Track_Length Column

A new column called **Revenue** was created in Dim_Track table using DAX in Power BI

This column categorizes the tracks by their length

- Short: shorter than 3 minutes
- Medium: between 3 and 7 minutes
- Long: longer than 7 minutes

```
Track_Length = SWITCH(TRUE(),
    Dim_Track[Milliseconds] < 180000 , "Short",
    Dim_Track[Milliseconds] >= 180000 && Dim_Track[Milliseconds] < 420000 , "Medium",
    Dim_Track[Milliseconds] >= 420000 , "Long")
```

Creation of the Date Dimension

A **date dimension table (DimDate)** was created using DAX to enable time-based analysis such as yearly trends, monthly comparisons, and year-over-year calculations.

Example DAX expression used to generate the date table:

Dim_Date = CALENDARAUTO()

The date table enables the use of **time intelligence functions** such as:

- Year-to-Date (YTD)
- Quarter-to-Date (QTD)
- Month-to-Date (MTD)
- Year-over-Year (YoY)

These calculations are essential for analyzing revenue trends over time.

Final Prepared Dataset

After applying these transformations, the dataset was structured into a clean analytical model consisting of:

- A central **fact table** containing transactional sales data
- Multiple **dimension tables** providing descriptive attributes for analysis
- A **date dimension** enabling advanced time-based analytics

This prepared dataset served as the foundation for building the interactive Power BI dashboard.

5. DAX MEASURES

Use of DAX in the Project

To support the analytical capabilities of the dashboard, several **DAX measures** were created in Power BI. These measures allow dynamic calculations that respond to filters and slicers within the dashboard.

The measures were used to calculate important business metrics such as revenue, customer value, and time-based performance indicators.

Total Revenue

The **Total Revenue** measure calculates the total income generated from track sales.

Total Revenue = SUM(invoiceline_Facttable[Revenue])

This measure is the main metric used across the dashboard to evaluate sales performance.

Total Orders

This measure counts the total number of invoices (orders).

Total Orders = DISTINCTCOUNT(invoice[InvoiceId])

It helps measure the number of transactions made by customers.

Number of Customers

This measure counts the total number of customers.

Number of Customers = DISTINCTCOUNT(Dim_Customer[CustomerId])

It provides an overview of the size of the customer base.

Average Order Value

This measure calculates the average value of each order.

Average Order Value = DIVIDE([Total Revenue],[Total Orders])

This metric helps evaluate how much customers spend per purchase.

Average Customer Value

This measure calculates the average revenue generated per customer.

Avg Revenue per Customer = DIVIDE([Total Revenue], [Number of Customers])

It provides insight into customer purchasing behavior.

Time Intelligence Measures

Time intelligence calculations allow the dashboard to analyze sales trends over time.

Year-to-Date Revenue (YTD)

YTD = TOTALYTD([Total Revenue],Dim_Date[Date])

This measure calculates revenue accumulated since the beginning of the current year.

Year-over-Year Revenue (YoY)

YoY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[Date]))

This measure calculates revenue from the same period in the previous year.

Growth Percentage

This measure calculates the percentage growth compared to the previous year.

Growth % = DIVIDE([Total Revenue] - [YoY], [YoY])

It helps identify whether revenue is increasing or decreasing over time.

Month-to-Date and Quarter-to-Date

Additional time-based measures were also created to support detailed time analysis.

MTD = TOTALMTD([Total Revenue],Dim_Date[Date])

QTD = TOTALQTD([Total Revenue],Dim_Date[Date])

These measures allow analysis of revenue within the current month or quarter.

Total Quantity

Total Quantity = SUM(invoiceline_Facttable[Quantity])

Album Count

Album Count = DISTINCTCOUNT(Dim_Track[Album])

Average Track Duration

Avg Track Duration = AVERAGE(Dim_Track[Milliseconds])/1000/60

Average Track Revenue

Avg Track Revenue = DIVIDE([Total Revenue],[Total Quantity])

Track Count

Track Count = COUNT(Dim_Track[TrackId])

Orders per Customer

Orders per Customer = DISTINCTCOUNT(invoice[InvoiceId])

Analytical Capabilities

The DAX measures enable the dashboard to dynamically analyze performance across multiple business dimensions such as:

- Artists and albums

- Music genres
- Customers and geographic regions
- Time periods such as months, quarters, and years

These calculations provide the analytical foundation for the interactive visualizations used throughout the dashboard.

6. DASHBOARD PAGES

Dashboard Design Approach

The Power BI dashboard was designed to provide a **clear and interactive analytical experience**. The visualizations were organized into multiple pages, with each page focusing on a specific analytical perspective of the business.

The design follows a **top-down analytical approach**, starting with high-level business performance indicators and progressively moving toward more detailed analyses such as artist performance, catalog structure, customer behavior, and time-based trends.

To improve usability, **interactive slicers and dynamic parameters** were implemented to allow users to filter and explore the data across different dimensions.

The final dashboard consists of **five analytical pages**.

Executive Overview

The **Executive Overview** page provides a high-level summary of business performance. It focuses on key metrics and major revenue drivers.

Key elements include:

- KPI cards displaying **Total Revenue, Total Orders, Total Customers, and Average Order Value**
- A **Top 10 Genres by Revenue** chart highlighting the most profitable music categories
- A **Top 10 Artists by Revenue** chart identifying the best-performing artists
- Geographic visualizations showing **Revenue by Country**
- Interactive slicers allowing filtering by **Year, Artist, Genre, and Country**

This page enables users to quickly understand the overall performance of the music store.

Artist Analysis

The **Artist Analysis** page focuses on evaluating artist and album performance.

Visualizations on this page include:

- KPI cards for **Total Revenue, Album Count, Tracks Sold, and Average Revenue per Track**
- A chart showing **artists with the largest catalog**, with a dynamic parameter allowing comparison by album count, track count, or average track duration
- A **Top 10 Artists** chart that can dynamically switch between revenue and quantity sold
- A **Top 10 Albums by Revenue** chart
- A table displaying **top-selling tracks** including track name, artist, album, and revenue

Slicers allow filtering by **Artist and Genre**, enabling deeper analysis of specific artists or music categories.

Music Catalog Analysis

The **Music Catalog** page explores the structure of the music library.

Key visualizations include:

- KPI cards for **Track Count, Genre Count, Average Track Duration, and Artist Count**
- A **Media Type distribution chart** that can dynamically switch between track count and revenue
- A **Track Length category chart** showing the distribution of track durations
- A **Genre revenue treemap** allowing users to explore the hierarchical contribution of genres
- A **Genre comparison chart** with a dynamic parameter to analyze genres by track count, album count, or average track duration

Slicers allow filtering by **Genre and Artist**, providing additional analytical flexibility.

Customer Analysis

The **Customer Analysis** page focuses on customer behavior and revenue contribution.

Key elements include:

- KPI cards displaying **Total Customers, Total Revenue, Average Customer Value, and Average Order Value**
- A chart analyzing **support representative performance**, with a dynamic parameter allowing comparison by revenue, quantity sold, or total orders
- A **Top 10 Customers by Revenue** visualization
- Geographic visualizations showing the **number of customers by country**

Slicers allow filtering by **Genre, Country, and Customer Name**.

Time Analysis

The **Time Analysis** page focuses on revenue trends and time-based performance.

Key visualizations include:

- KPI cards displaying **Total Revenue, Year-over-Year Revenue, Year-to-Date Revenue, and Growth Percentage**
- A **Revenue Trend line chart** showing revenue across the date hierarchy
- A chart comparing **Growth Percentage and Year-over-Year revenue over time**
- A **matrix visualization** displaying revenue metrics across different time levels (year, quarter, month, and day)

This page enables users to analyze business performance over time and identify trends or seasonal patterns.

Interactive Features

To enhance user experience, the dashboard includes several interactive features:

• Slicers for filtering data by year, genre, artist, country, and customer
• Dynamic field parameters allowing users to switch between different metrics
• Cross-filtering between visualizations
• Hierarchical drill-down capabilities in time-based charts
• Buttons to navigate between pages and switch between light and dark modes

These features allow users to explore the data dynamically and gain deeper insights from the dashboard.